BEGINNING OF TRANSMISSION

State: published
Pairing: complete (DOI, GitHub, SHA-256, Zenodo URL)

Release metadata

Field	Value
Title	A template/ approach to Reproducible Generative Research
Version	1.0.9
Concept DOI	10.5281/zenodo.20419007
Version DOI	10.5281/zenodo.20976048
GitHub	https://github.com/docxology/template_template/releases/tag/v1.0.9
Zenodo	https://zenodo.org/records/20419007
SHA-256	`535bd80943d0ae9f…`
SHA-512	pending

How to verify

- Scan **Integrity** QR and compare the embedded SHA-256 prefix to the table above.
- Scan **Zenodo** / **GitHub** QR codes and confirm they resolve to this release pairing.
- Full hashes and structured fields: `../data/transmission_manifest.json`.

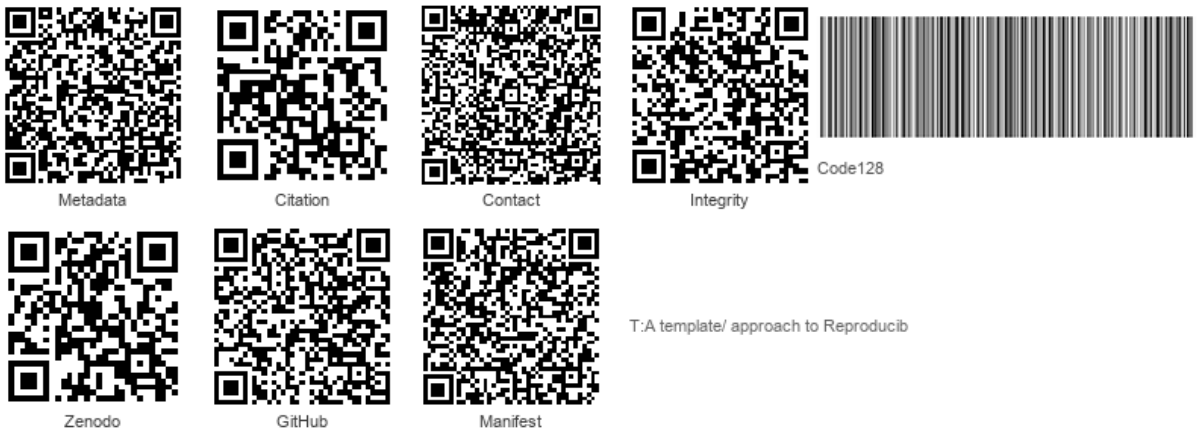

Figure 1: Integrity QR strip

Structured manifest: `../data/transmission_manifest.json`

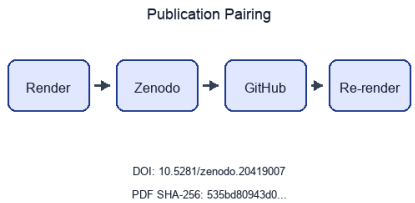

Figure 2: Publication pairing flow

Stego: off | overlays text | barcodes on | XMP on | manifest on → `./secure_run.sh`

A template/ approach to Reproducible Generative Research

Architecture and Ergonomics from Configuration through Publication

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Contents

1 Abstract

The reproducibility crisis in computational research is fundamentally structural: research artifacts are scattered across disconnected tools—LaTeX editors, Jupyter notebooks, ad-hoc shell scripts—with no enforced mechanism to keep code, data, and manuscript synchronized. Studies have shown that most published findings are false positives, replication rates in psychology hover around 36%, and only 24% of 1.4 million Jupyter notebooks can be successfully re-executed. Existing tools address fragments of this problem: workflow managers (Snakemake, Nextflow, CWL) orchestrate computation; literate programming systems (Quarto, Jupyter Book, R Markdown, Overleaf, OpenAI Prism) render documents; data versioning tools (DVC) track artifacts—but none enforces cross-cutting quality standards as architectural invariants. `template/` applies the principle of Infrastructure as Code to the research lifecycle, making the manuscript, test suite, and provenance chain version-controlled, deterministically buildable, and independently verifiable. It is built on a Two-Layer Architecture that separates 23 infrastructure subdirectories (20 importable Python packages, about 604 modules, validated by about 7,904 tests) from self-contained project workspaces, connected by a YAML-declared pipeline (12 stages; default full 10)-based build pipeline progressing from environment sanitization through test execution (with a Zero-Mock testing policy enforcing 90% project-level and 60% infrastructure-level coverage via real filesystem operations and subprocess invocations), analysis script invocation, Pandoc/XeLaTeX rendering, SHA-256 cryptographic hashing with steganographic watermarking, structural PDF validation, and LLM-assisted review. A Documentation Duality standard equips every directory with both human-readable `README.md` and machine-readable `AGENTS.md` files, while each infrastructure module additionally carries a `SKILL.md`—a structured skill descriptor aligned with the Model Context Protocol—enabling AI agents to locate and invoke module capabilities without hallucinating API signatures. Scalability is demonstrated across the generated public exemplar roster (`templates/template_active_inference`, `templates/template_autoresearch_project`, `templates/template_autoscientists`, `templates/template_code_project`, `templates/template_eda_notebook`, `templates/template_gold_refinement`, `templates/template_literature_meta_analysis`, `templates/template_madlib`, `templates/template_methods_paper`, `templates/template_newspaper`, `templates/template_prose_project`, `templates/template_search_project`, `templates/template_sia`, `templates/template_template`, `templates/template_textbook`), with representative heterogeneous cases under `projects/templates/`: optimization (`template_code_project`, 236 tests), prose (`template_prose_project`, 120 tests), and AutoResearch readiness (`template_autoresearch_project`, 296 tests). These guarantee control-positive layouts for code-centric, prose-centric, and retrieval-centric workflows at 90%+ project coverage alongside 60%+ infrastructure gates. All three share identical pipeline stages without cross-project coupling. This manuscript adds a complementary reflexive artifact: authored from `projects/templates/template_template` (130 tests) as a public exemplar in the same discovered/rendered tree, using the same analysis and render path and injecting counters from repository introspection. The fact that these words, metrics, and figures were generated by the pipeline they describe demonstrates self-documenting capacity: rendered through the DAG, validated without mocks, optionally watermarked. A comparative analysis against nine peer tools across fourteen dimensions positions `template/` as integrating fourteen distinctive enforcement capabilities—testing thresholds, cryptographic provenance, steganographic watermarking, multi-project management, MCP-aligned skill descriptors, Zero-Mock policy, orchestration through publication—in one repository. Code is released under the Apache License 2.0 at `github.com/docxology/template`; the work remains open-ended.

2 Introduction

2.1 Research Tools as Epistemic Infrastructure

Scientific research operates through a layered ecology of tools, documents, and practices—each shaping what can be known, communicated, and verified. When these layers are fragmented, the artifacts of research (manuscripts, data, code, figures) drift out of alignment with one another, creating gaps that are structural rather than incidental. The "reproducibility crisis" is one symptom of this deeper misalignment: a 2016 *Nature* survey of 1,576 researchers found that 70% had tried and failed to reproduce another scientist's experiments, and more than half had failed to reproduce their own [Baker, 2016]. Freedman et al. estimate that the biomedical industry alone loses $28 billion annually to irreproducible preclinical research [Freedman et al., 2015]. But reproducibility is only the most visible face of a broader problem. Research software engineering, epistemic integrity, and the coordination between human researchers and AI collaborators all depend on the same underlying question: whether the tools that produce research artifacts can themselves be made transparent, testable, and self-documenting. Nosek et al. [Nosek et al., 2018] have argued that the preregistration revolution—requiring researchers to commit to analytical plans before data collection—is a necessary structural reform; we extend this logic to the entire research build pipeline.

One root cause is fragmentation—of attention (in the mind) as well as in the cyberphysical niche (documents, versions, and reproducibility artefacts). A typical research project scatters its artifacts across disconnected tools: LaTeX editors [Lamport, 1994] for prose, Jupyter notebooks for analysis, ad-hoc shell scripts for figure generation, and manual copy-paste for integrating results into manuscripts. Each boundary between tools is a potential locus of desynchronization. The version of the figure embedded in the PDF may not match the version of the code that ostensibly generated it. The test suite, if it exists at all, likely tests the code in isolation from the rendering pipeline. Pimentel et al. [Pimentel et al., 2019] analyzed 1.4 million Jupyter notebooks from GitHub and found that only 24% could be successfully re-executed, with 36% producing different results—quantifying the reproducibility cost of notebook-based workflows. Peng [Peng, 2011] argues that reproducibility in computational science requires, at minimum, that the data and code underlying a published result be available for independent verification—yet the tools for enforcing this standard remain ad hoc. Indeed, even the terminology is fractured: Barba [Barba, 2018] documents how "reproducibility," "replicability," and "repeatability" carry conflicting definitions across disciplines, undermining cross-field standards.

2.2 Related Work

Gentleman and Temple Lang [Gentleman and Temple Lang, 2007] introduced the concept of a *research compendium*—a single unit of scholarly communication bundling code, data, and narrative. This vision has driven two decades of tooling, which can be grouped into four categories: workflow managers, literate programming systems, containerization approaches, and best-practice frameworks.

2.2.1 Workflow Managers

Snakemake [Köster and Rahmann, 2012] uses a rule-based, Python-derived DSL to specify computational workflows as directed acyclic graphs of file-producing steps. It supports containerized execution via Conda and Singularity environments. Snakemake 9.x (2024–2025) introduced a plugin architecture for extended execution backends and storage providers, yet its scope remains computational pipeline orchestration—it does not integrate manuscript rendering, testing enforcement, or provenance watermarking.

Nextflow [Di Tommaso et al., 2017] employs a dataflow programming paradigm with native support for container-based execution across heterogeneous computing environments (local, SLURM, AWS). Like Snakemake, Nextflow excels at bioinformatics pipeline parallelism but does not address manuscript production, document integrity, or the testing–publication coupling that characterizes research reproducibility.

CWL (Common Workflow Language) [Amstutz et al., 2016] provides a portable, YAML-based standard for describing computational workflows and their dependencies. Its strength lies in interoperability across execution engines (cwltool, Toil, Arvados), but it requires external tooling for manuscript generation and offers no built-in testing or provenance framework.

2.2.2 Literate Programming and Publication Tools

Knuth's literate programming [Knuth, 1984] established the principle that programs should be authored as documents intended for human comprehension. Schulte et al. [Schulte et al., 2012] extended this to multi-language computing environments (Org-mode), demonstrating that literate programming could span languages and output formats.

Quarto [Allaire et al., 2024] extends the R Markdown tradition to support Python, Julia, and Observable, rendering to PDF, HTML, and Word. Quarto integrates code execution with document rendering, achieving a modern form of literate programming, but it does not enforce testing thresholds, manage multi-project repositories, or provide cryptographic provenance.

Jupyter Book [Kluyver et al., 2016] builds on Jupyter notebooks to produce publication-quality documents via Sphinx. While powerful for interactive exploration, Jupyter's notebook format introduces execution-order fragility [Pimentel et al., 2019] and does not naturally support the separation of logic from orchestration that characterizes maintainable research software.

R Markdown [Xie et al., 2018] pioneered knitr-based dynamic documents that weave code and prose. Its ecosystem is rich but R-centric, and it lacks the multi-project management, infrastructure testing, and provenance embedding that characterize `template/`.

Typst [Mädje and Haug, 2023] is an emerging markup-based typesetting system with incremental compilation and a programmable scripting layer. While Typst offers faster compilation than LaTeX and a more modern authoring experience, it does not integrate testing, provenance, or multi-project pipeline management.

2.2.3 Containerization and Environment Capture

Docker [Boettiger, 2015] addresses reproducibility at the environment level—packaging operating system, libraries, and code into portable containers. While Docker solves the "works on my machine" problem, containerization is complementary to, not a replacement for, the architectural concerns addressed here: Docker does not enforce testing, embed provenance, or manage multi-project manuscript workflows.

2.2.4 Best-Practice Frameworks and Data Standards

Wilson et al. [Wilson et al., 2017] define "good enough" practices for scientific computing, emphasizing version control, testing, and documentation. Sandve et al. [Sandve et al., 2013] propose ten rules for reproducible computational research. Piccolo and Frampton [Piccolo and Frampton, 2016] systematically survey tools for computational reproducibility, finding that environment isolation, workflow automation, and documentation generation address complementary but non-overlapping reproducibility concerns—yet no single tool unifies all three. Stodden et al. [Stodden et al., 2016] advocate for enhanced computational method transparency. The FAIR principles [Wilkinson et al., 2016]—Findable, Accessible, Interoperable, Reusable—establish a standard for data stewardship that has been widely adopted by funding agencies and journals. Lamprecht et al. [Lamprecht et al., 2020] formalize "Towards FAIR Principles for Research Software," providing the conceptual scaffolding that Barker et al. [Barker et al., 2022] would soon operationalize as the FAIR4RS initiative—recognizing that software has execution, composability, and dependency-management requirements that data-centric FAIR does not address. Cohen et al. [Cohen et al., 2021] characterize the four pillars of research software engineering (software sustainability, software quality, community building, and policy advocacy), situating formal testing and provenance practices within a broader RSE governance framework. Garijo et al. [Garijo et al., 2024] operationalize FAIR4RS through the FAIRsoft evaluator, an automated assessment framework that scores research software against 17+ quality indicators including executability, metadata richness, and documentation completeness. Goble et al. [Goble et al., 2020] extend FAIR to computational workflows specifically, arguing that workflow provenance requires first-class treatment in scientific computing infrastructure. Nüst et al. [Nüst et al., 2017] introduce the *executable research compendium* (ERC), extending Gentleman and Temple Lang's compendium concept with containerized, interactive reproduction environments. The W3C PROV data model [Moreau and Missier, 2013] provides a formal vocabulary for expressing provenance records, while in-toto [Torres-Arias et al., 2019] provides a framework for end-to-end software supply chain integrity verification, and SLSA [Open Source Security Foundation, 2023] (Supply-chain Levels for Software Artifacts) extends this to graduated, attestation-based supply-chain

security levels for build pipelines. These frameworks articulate *what* reproducible research requires but do not provide an integrated *how*—they lack the tooling, enforcement mechanisms, and architectural patterns that translate standards into practice.

2.2.5 The Gap

Despite advances in FAIR4RS principles [Barker et al., 2022, Lamprecht et al., 2020], automated FAIR software assessment [Garijo et al., 2024], FAIR computational workflow standards [Goble et al., 2020], supply-chain attestation frameworks [Torres-Arias et al., 2019, Open Source Security Foundation, 2023], and the preregistration revolution [Nosek et al., 2018], no existing system integrates six cross-cutting concerns into a single enforced pipeline: (1) end-to-end pipeline orchestration with testing enforcement, (2) multi-format manuscript rendering, (3) cryptographic provenance embedding, (4) multi-project repository management, (5) FAIR-aligned software stewardship, and (6) AI-agent collaboration via structured documentation. Each existing framework addresses a subset; none provides the unified enforcement mechanism. The detailed tool-by-tool comparison is developed in the Comparison to Existing Tools section; the summary table below captures the gap landscape.

2.2.6 Summary: Gaps Left by Existing Tools

The six concerns identified above map onto existing tool categories as follows:

Gap	Partially Addressed By	Not Addressed By
Pipeline orchestration	Snakemake 9.x, Nextflow 25.x, CWL 1.2	Quarto, Jupyter Book, R Markdown, Typst, Overleaf, Prism
Manuscript rendering	Quarto 1.x, Jupyter Book 2.x, R Markdown, Typst, Overleaf (2025), Prism	Snakemake, Nextflow, CWL, DVC
Testing enforcement	—	All existing tools
Cryptographic provenance	SLSA (build-level attestation only)	All research-focused tools
Multi-project management	—	All existing tools
AI-agent documentation	Overleaf (partial co-author AI), Prism (partial context reasoning)	All pipeline/workflow tools
Agentic skill protocol (MCP-aligned)	—	All existing tools

No existing system addresses all six concerns within a single enforced pipeline.

2.3 `template/`: An Integrated Solution

`template/` was conceived as a structural antidote to this fragmentation. Rather than adding reproducibility as an afterthought—a Docker container wrapping an already-disjointed workflow [Boettiger, 2015]—the template enforces integrity at the architectural level. It realizes Gentleman and Temple Lang's research compendium vision [Gentleman and Temple Lang, 2007] at repository scale, bundling code, data, tests, manuscripts, and provenance into a single, pipeline-enforced system with version-controlled infrastructure [Ram, 2013]. It stands on four primary pillars:

1. **Ergonomic Modularity**: A Two-Layer Architecture cleanly separates globally shared infrastructure (logging, rendering, validation, steganography) from project-specific logic (manuscripts, scripts, data). 23 infrastructure subdirectories (20 importable packages) comprising about 604 Python modules provide reusable services; projects consume them without modification.
2. **Execution Integrity**: A Zero-Mock testing policy where pipeline advancement is contingent on test passage. Infrastructure tests must achieve 60% coverage; project tests must achieve 90%. All tests use real filesystem operations, real subprocess calls, and real network connections—no mock objects,

no fake services, no synthetic test doubles. about 7,904 infrastructure tests and 3377+ project tests enforce this standard.

3. **Automated Provenance**: Steganographic watermarking and cryptographic hashing are integrated directly into the rendering pipeline. Every generated PDF carries a SHA-256 fingerprint, an alpha-channel text overlay encoding the build timestamp and commit hash, and optionally a QR code linking to the repository. Provenance is not asserted by policy; it is enforced by architecture.
4. **AI-Agent Collaboration and Skill-Based Agentic Operations**: A three-tier documentation architecture enables AI agents to operate at every level of the system. At the system level, `CLAUDE.md` asserts global architectural constraints. At the structural level, `AGENTS.md` files at every directory expose local API surfaces, file inventories, and integration contracts. At the module level, `SKILL.md` files—written to a discoverable YAML+Markdown schema aligned with the Model Context Protocol [Anthropic, 2024]—define each infrastructure module as a reusable, self-describing tool. An agent invoking `infrastructure.rendering` does not need to read source code: it reads the `rendering/SKILL.md`, which declares the module's name, description, key imports, and example invocations in a machine-parseable YAML frontmatter block. This architecture is the practical realization of the skill-library paradigm established in the agent literature: Yao et al.'s ReAct framework [Yao et al., 2023] demonstrated that interleaving reasoning traces with tool invocations dramatically improves LLM reliability; Schick et al.'s Toolformer [Schick et al., 2023] showed that self-supervised tool use can be bootstrapped from natural language; Wang et al.'s Voyager [Wang et al., 2023] proved that growing skill libraries enable open-ended autonomous exploration in complex environments. `template/` instantiates this vision in the domain of scientific research infrastructure: each `SKILL.md` is a Voyager-style skill, each pipeline stage is a ReAct action, and the full infrastructure layer constitutes a composable, protocol-aligned skill library for scientific computation.

2.4 Scope and Contributions

This paper is itself a product of the template it describes. The metrics populating its tables were computed by the introspection module documented in the Methods; the figures were rendered by the visualization code validated by the test suite described in Quality Assurance; the PDF carrying these words was assembled by the same YAML-declared pipeline whose architecture is the subject of the Results. This self-productive loop—where the system that is described is also the system that produces the description—is not incidental but structural, a concrete demonstration that `template/` can sustain the full lifecycle from source code to published artifact within a single, version-controlled, pipeline-enforced repository. Our contributions are:

- A formal description of the Two-Layer Architecture and Standalone Project Paradigm that enables N independent research projects to share infrastructure without coupling.
- A detailed specification of the 12-stage DAG in `infrastructure/core/pipeline/pipeline.yaml`: default full runs use 10 stages; `--core-only` runs 8; opt-in bundle and archival stages via `--tags`.
- A comparative analysis positioning `template/` against nine peer tools—Snakemake, Nextflow, CWL, Quarto, Jupyter Book, R Markdown, DVC, Overleaf, OpenAI Prism—across fourteen feature dimensions, demonstrating that `template/` uniquely bundles the fourteen enforcement capabilities enumerated in §Results.
- An empirical evaluation across the canonical exemplar projects under `projects/templates/` (`templates/template_active_inference`, `templates/template_autoresearch_project`, `templates/template_autoscientists`, `templates/template_code_project`, `templates/template_eda_notebook`, `templates/template_gold_refinement`, `templates/template_literature_meta_analysis`, `templates/template_madlib`, `templates/template_methods_paper`, `templates/template_newspaper`, `templates/template_prose_project`, `templates/template_search_project`, `templates/template_sia`, `templates/template_template`, `templates/template_textbook`)—including this meta manuscript from `projects/templates/template_template`, which exercises introspection-derived metrics.
- A security analysis of the steganographic provenance layer, including a formal threat model and tamper-detection capabilities aligned with the W3C PROV data model [Moreau and Missier, 2013] and SLSA [Open Source Security Foundation, 2023].
- An open-source reference implementation available at `github.com/docxology/template`.

2.5 Paper Organization

The Methods describe the Two-Layer Architecture, Thin Orchestrator pattern, pipeline stages, and AI collaboration model. Results present quantitative metrics from multi-project execution, coverage analysis, and steganographic benchmarks. The Discussion addresses the Zero-Mock tradeoff, scalability implications, a detailed tool comparison, and future directions. The Infrastructure Module Reference provides detailed documentation for all 23 subdirectories. Security and Provenance describes the steganographic and cryptographic integrity layer. The Appendices provide pipeline, configuration, and comparative references.

3 Methods

The `template/` architecture is deliberately bifurcated into a globally shared `infrastructure/` layer and project-specific `projects/` silos. This section describes the four core design patterns, the YAML-declared pipeline (12 stages; default full 10) pipeline that operationalizes them, and the AI collaboration model that distinguishes this system from conventional research templates.

3.1 The Two-Layer Architecture

The repository is organized into two strictly separated layers:
Infrastructure Layer (`infrastructure/`): 23 infrastructure subdirectories—20 of them independently-importable Python packages—comprising about 604 modules and providing reusable services. Each importable package has its own `__init__.py`, `AGENTS.md`, and `README.md`, and exports a well-defined public API (the remaining subdirectories, e.g. `config/`, hold shared configuration). The infrastructure layer knows nothing about any specific project—it provides generic capabilities (logging, rendering, validation, steganography) that any project may consume.
Project Layer (`projects/`): Self-contained research workspaces. Each project directory contains:

Directory	Purpose
`manuscript/`	Markdown chapters and `config.yaml`
`scripts/`	Thin orchestrator scripts (Stage 02)
`src/`	Project-specific Python modules
`tests/`	Project-specific test suite
`data/`	Input datasets and generated data
`output/`	Pipeline artifacts: PDF, figures, reports, logs
`docs/`	Project-specific architecture documentation

The two layers communicate exclusively through Python imports and filesystem paths. No project modifies infrastructure code; no infrastructure module references a specific project by name (except via runtime project discovery).

3.2 The Standalone Project Paradigm

Projects are designed to be completely self-contained. Adding a new project requires no changes to the infrastructure layer, no modifications to `pyproject.toml`, and no updates to the pipeline orchestrator. A project is automatically discovered if and only if it satisfies two conditions:

1. It exists as a subdirectory of `projects/`.
2. It contains the file `manuscript/config.yaml`.

This paradigm enables horizontal scaling: N researchers can maintain N independent projects within a single repository, sharing infrastructure without coupling. Each project declares its own testing tolerances, manuscript metadata, LLM review preferences, and rendering configuration in its `config.yaml`. The system currently hosts its public canonical exemplars under `projects/templates/` (`templates/template_active_inference`, `templates/template_autoresearch_project`, `templates/template_autoscientists`, `templates/template_code_project`, `templates/template_eda_notebook`, `templates/template_gold_refinement`, `templates/template_literature_meta_analysis`, `templates/template_madlib`, `templates/template_methods_paper`, `templates/template_newspaper`, `templates/template_prose_project`, `templates/template_search_project`, `templates/template_sia`, `templates/template_template`, `templates/template_textbook`), including this meta-manuscript at `projects/templates/template_template/`.

3.3 The Thin Orchestrator Pattern

All scripts in `scripts/` (both infrastructure-level and project-level) follow the Thin Orchestrator pattern [Gamma et al., 1995]:

- **No domain logic**: Scripts contain zero algorithmic implementation. They import functions from `src/` modules and wire them to infrastructure services.
- **Configuration-driven**: Behavior is parameterized by `config.yaml`, not by hardcoded values.
- **Stateless**: Scripts read inputs, call functions, write outputs. They maintain no persistent state between invocations.
- **Logged**: Every significant action is logged via `infrastructure.core.logging.utils.get_logger`.

This pattern ensures that all testable logic lives in `src/` where it is subject to the Zero-Mock testing policy, while scripts remain thin enough to be audited by visual inspection. The separation draws on the Mediator pattern from Gamma et al. [Gamma et al., 1995], where scripts mediate between infrastructure services and project-specific code without implementing any logic of their own.

To make this concrete, the following contrasts the anti-pattern with the correct pattern:

```
# ANTI-PATTERN: domain logic embedded in script
def calculate_average(data):          # ← never put computation here
    return sum(data) / len(data)

result = calculate_average([1, 2, 3])

# CORRECT PATTERN: script imports from src/ and only wires
from projects.my_project.src.statistics import calculate_average

result = calculate_average([1, 2, 3])  # ← scripts wire, never compute
```

The critical property is that `calculate_average` in the correct pattern lives in a testable `src/` module, is covered by the Zero-Mock test suite, and can be independently imported, tested, and reused—whereas the anti-pattern buries logic in a script that is invisible to coverage tools.

3.4 DAG Pipeline Declared by `pipeline.yaml`

Single-project pipelines read `infrastructure/core/pipeline/pipeline.yaml`. `scripts/execute_pipeline.py` expands the declarative DAG, applies tag filters (`--core-only` skips `llm` stages), checkpoints between nodes, then dispatches numbered scripts (`scripts/NN_*.py`) or builtin methods (`_run_clean_outputs`).
The **default YAML graph contains ten named stages** (plus telemetry configuration metadata):

1. **Clean Output Directories** — wipes prior `projects/<name>/output/` + delivered `output/<name>/` paths so stale PDFs cannot satisfy validation.
2. **Environment Setup** (`00_setup_environment.py`) — Python/uv probing, toolchain discovery, scaffolding directories, `PYTHONPATH` wiring.
3. **Infrastructure Tests** (`01_run_tests.py --infra-only`) — `tests/` suite with infra coverage thresholds (>=60 %).
4. **Project Tests** (`01_run_tests.py --project-only`) — per-project suites with >=90 % coverage mandate.
5. **Project Analysis** (`02_run_analysis.py`) — lexicographically ordered `projects/<name>/scripts/*.py`, each a thin orchestrator (`src/` does real work).
6. **PDF Rendering** (`03_render_pdf.py`) — Pandoc → XeLaTeX loop, bibliography assembly, injected variables from Stage 02 artefacts.
7. **Output Validation** (`04_validate_output.py`) — PDF structure, manifests, Markdown hygiene.
8. **LLM Scientific Review** (`06_llm_review.py --reviews-only`; `tags: llm`) — executive + quality critiques via local Ollama; `allow_skip: true`.
9. **LLM Translations** (`06_llm_review.py --translations-only`; tags `llm`, same dependency edges) — multilingual abstract expansion.
10. **Copy Outputs** (`05_copy_outputs.py`) — reproducible snapshots into canonical `output/<project>/`.

Two LLM nodes intentionally share one script module with orthogonal CLI switches; both depend only on validation so they can parallelize logically while remaining optional.
Executive reporting (`scripts/07_generate_executive_report.py`) is **not** a YAML node inside the single-project executor. `--all-projects` / `execute_multi_project.py` invokes it once after iterating projects, consolidating cross-project KPIs dashboards.
Topological order therefore differs slightly from lexical script numbering (e.g., copy executes after validation even though script `05` precedes `06` lexically).

3.4.1 Stage Highlights

Infrastructure vs project tests. Splitting pytest invocations isolates flaky infra regressions (`MAX_TEST_FAILURES` knobs) from zero-tolerance gates on domain code (`max_project_test_failures` default 0 declared in YAML front-matter/testing blocks).
Stage 02 illustration. The analysis stage is deliberately concrete rather than a hypothetical diagram factory: each canonical project ships real behaviour at this node. `template_autoresearch_project` runs readiness validation; `template_search_project` merges remote literature JSON, generates scripted figures (`y_generate_search_figures.py`), and writes manifests; `template_code_project` emits optimization plots; and `template_prose_project` triggers structural validation scaffolding. The pipeline shape is identical across all four—only the Stage 02 payload differs—which is exactly what lets one orchestrator serve heterogeneous research domains.

3.4.2 Interactive Orchestration

3.4.2.1 `run.sh` Thin wrapper invoking `python -m infrastructure.orchestration`. Offers:

- per-project staged execution,
- chained digits (`234` shorthand),
- multi-project grid (`a`–`d` presets),
- graceful quit / resume parity with `scripts/README.md`.

Selecting **`d` alone** after a passing multi-project run exits immediately once summaries print—avoiding repetitive menu redraw.

3.4.2.2 `secure_run.sh` Executes Python `secure` path: standard pipeline artefact reproduction **then** invokes `run_secure_pipeline` for steganographic PDF hardening (`infrastructure.steganography`). Original PDFs stay immutable; hardened companions carry QR overlays plus hash manifests sidecars.

3.5 Documentation Duality and AI Collaboration

Every directory at every level of the repository hierarchy contains two documentation files:

- **README.md**: Human-readable overview, quick-start instructions, and directory structure.
- **AGENTS.md**: Machine-readable technical specification optimized for AI coding assistants. Contains API tables, dependency graphs, implementation patterns, and architectural constraints.

This Documentation Duality standard serves two purposes. First, it ensures that both human researchers and AI agents can navigate the codebase efficiently—AGENTS.md files provide the structured context that language models need to make informed code modifications without hallucinating API signatures or violating architectural invariants. Second, it creates a self-documenting feedback loop: as AI agents modify the codebase, they update the corresponding AGENTS.md files, keeping documentation synchronized with implementation. Lau and Guo's survey of 90 AI coding assistant systems [Lau and Guo, 2025] identifies contextual code understanding as a primary bottleneck; the Documentation Duality standard addresses this by providing pre-structured context at every directory level.

The template additionally includes CLAUDE.md at the repository root, providing system-level instructions for AI coding assistants—architectural principles, testing requirements, and naming conventions that apply globally. This creates a three-tier documentation architecture: per-directory AGENTS.md for local context, root README.md and CLAUDE.md for global constraints, and README.md for human navigation.

3.6 Agentic Skill Architecture

The Documentation Duality standard addresses human and AI navigation at the directory level. A complementary layer operates at the *module* level: every infrastructure subpackage carries two additional machine-readable files that transform it from a passive code library into an active, protocol-aligned skill endpoint.

3.6.1 The Three-Tier Skill Protocol

template/ implements a progression of agent-facing documentation, escalating in specificity from global constraints to module-level API contracts:

Tier	File	Scope	Purpose
1 — System	README.md	Repository root	Global architectural principles, Zero-Mock policy, naming conventions
2 — Structure	AGENTS.md	Every directory	Local file inventories, API surfaces, integration patterns, architectural constraints
3 — Skill	SKILL.md	Every infrastructure module	Machine-parseable skill descriptor: module name, description, key imports, usage pattern

Tier 1 and Tier 2 have direct analogues in the prompt-engineering literature: system prompts and retrieval-augmented context [Lau and Guo, 2025]. Tier 3 is novel. The SKILL.md files follow a YAML frontmatter + Markdown instruction format precisely aligned with the tool-descriptor schemas of the Model Context Protocol [Anthropic, 2024]. The following is the exact frontmatter from infrastructure/rendering/SKILL.md:

```yaml
---
name: rendering
description: >
  Multi-format output generation (PDF, HTML, slides).
  Use for: Pandoc/XeLaTeX rendering, RenderManager, slide deck generation.
  Key imports: RenderManager, RenderingConfig from infrastructure.rendering
---
```

An MCP client reading this block immediately knows the module name, its natural-language activation condition ("use for"), and which Python symbols to import. No source-code inspection is required. This is the practical implementation of Toolformer-style self-documented tools [Schick et al., 2023]—rather than a language model learning tool APIs from training data, the APIs are encoded directly in version-controlled, co-located skill files that evolve with the codebase.

3.6.2 Module Skill Coverage

Each infrastructure subdirectory surfaced by `discover_infrastructure_modules()` carries paired `README.md` + `AGENTS.md`; agent-facing `SKILL.md` manifests exist wherever teams enable Cursor / PAI manifests (regenerated via `python -m infrastructure.skills`). Root-level `PAI.md` summarizes cross-package obligations.
Promotion policy: new Layer-1 directories must ship human + machine-readable docs (`README.md`, `AGENTS.md`) immediately; Tier-3 SKILL assets follow once APIs stabilize.

3.6.3 MCP Server Mapping

The mapping from `SKILL.md` descriptors to MCP server endpoints is intentional but not yet automated; it represents the principal next integration step. In the MCP architecture [Anthropic, 2024], a server exposes three primitive types: **Tools** (executable functions), **Resources** (data the model can read), and **Prompts** (structured query templates). Each `infrastructure` module maps naturally onto this taxonomy:

- `infrastructure.llm` → MCP **Tool** (`query`, `apply_template`) + MCP **Prompt** (research prompt templates)
- `infrastructure.rendering` → MCP **Tool** (`render_pdf`, `render_html`) + MCP **Resource** (rendered PDFs as URI-addressable resources)
- `infrastructure.validation` → MCP **Tool** (`validate_pdf_rendering`, `validate_markdown`)
- `infrastructure.publishing` → MCP **Tool** (`publish_to_zenodo`, `generate_citation_bibtex`) + MCP **Resource** (DOI registry)
- `infrastructure.steganography` → MCP **Tool** (`SteganographyProcessor.process`) + MCP **Resource** (hash manifests)
- `infrastructure.search` · `infrastructure.reference` → MCP **Tool** wrappers over literature retrieval + BibTeX handling + MCP **Resource** exports for corpus JSON / `.bib`

An agent orchestrating a full research pipeline could, in principle, compose these MCP tools to reproduce the declarative DAG programmatically—discovering capabilities via `SKILL.md` frontmatter, executing them via MCP protocol calls, and consuming their outputs as Resources. The `SKILL.md` files parallel Voyager's skill library [Wang et al., 2023]—Voyager's agent accumulates a growing library of executable Minecraft skills represented as JavaScript functions; `template/`'s agent accumulates a curated library of research pipeline skills represented as YAML-frontmattered `SKILL.md` files. In both cases, the skill representation is machine-readable, version-controlled, and self-describing. Wang et al.'s LLM agent survey [Wang et al., 2024a] identifies tool use, planning, and memory as the three fundamental capabilities of autonomous agents; Yao et al.'s ReAct framework [Yao et al., 2023] demonstrates that interleaving reasoning traces with tool actions dramatically improves agent reliability in interactive settings. The `template/` skill architecture maps cleanly onto these three capabilities: the `SKILL.md` descriptors supply the tool-use layer, the declarative DAG of `12` `pipeline.yaml` stages (a default full run executes `10`) supplies the planning scaffold, and the per-criterion checkpoint system supplies the memory layer.

3.7 FAIR Alignment and Research Infrastructure as Code

The template's design aligns with both the original FAIR principles [Wilkinson et al., 2016] and the FAIR for Research Software (FAIR4RS) principles [Barker et al., 2022] at the repository level. FAIR4RS recognizes that software has requirements distinct from data—executability, composability, and dependency management—and the template addresses each.

3.7.1 Principle-by-Principle Alignment

Findability. Outputs are *Findable* through standardized directory structures, manifest files, and machine-readable metadata embedded in PDFs. Every project's `config.yaml` provides structured metadata (title, authors, DOIs, keywords) in a format parseable by both Pandoc and external indexing services. The `metrics.json` output provides a machine-readable inventory of all generated artifacts, their locations, and their provenance hashes.

Accessibility. Outputs are *Accessible* via open-source distribution on GitHub, with metadata embedded in the artifact itself rather than in a separate registry. The steganographic layer embeds provenance information directly in the PDF—including SHA-256 content hashes, build timestamps, and QR-encoded metadata—ensuring accessibility even when the PDF circulates outside the repository.

Interoperability. *Interoperability* is achieved through standard formats (PDF, JSON, BibTeX, YAML) and well-defined module APIs that enable cross-project composition. The Pandoc rendering pipeline accepts any Markdown-with-LaTeX input conforming to the template's section numbering conventions, allowing seamless migration of manuscripts from other Pandoc-based workflows.

Reusability. *Reusability* is ensured by the Standalone Project Paradigm—any project can be extracted and reused independently—and by the Documentation Duality standard, which satisfies FAIRsoft's inspectability and documentation quality indicators [Garijo et al., 2024]. The pipeline's automated testing and coverage enforcement directly operationalize the FAIR4RS executability requirement: software that cannot pass its own test suite cannot produce publishable output.

3.7.2 Infrastructure as Code for Research

At a higher level of abstraction, `template/` applies the DevOps principle of *Infrastructure as Code* (IaC) to the research lifecycle. In production software engineering, IaC means that server configuration is version-controlled, automatically provisioned, and independently reproducible [Wilson et al., 2017]. `template/` extends this principle to the research manuscript: the document is not authored in a word processor and emailed to collaborators, but *built* from version-controlled Markdown sources, *tested* against formal coverage thresholds, and *deployed* to a provenance-embedded PDF.

Every component of the research pipeline—the test suite, the analysis scripts, the rendering configuration, and the steganographic watermark—is specified in code, committed to git, and reproducible from a clean checkout. This deterministic build property means that any researcher can clone the repository, run `./run.sh --pipeline`, and produce a byte-for-byte identical manuscript (modulo timestamps in the steganographic metadata).

Software Heritage [Di Cosmo et al., 2020] provides persistent SWHIDs (Software Hash Identifiers) for source code snapshots, enabling stable citation of any specific version of `template/` as a discrete software artifact—closing the loop from research infrastructure to citable scientific contribution. Combined with Zenodo DOI registration (supported by `infrastructure.publishing`), this creates a dual-identifier citation chain: SWHID for source provenance, DOI for publication metadata [Katz et al., 2021].

3.8 Quality Assurance

3.8.1 Zero-Mock Testing Policy

All tests use real methods exclusively [Martin, 2008, Meszaros, 2007]. No `unittest.mock`, no `MagicMock`, no `patch` decorators. Tests that require external services (Ollama, network) use `pytest.mark` markers for conditional execution. The philosophical motivation—analogizing mock objects to Simmons et al.'s *researcher degrees of freedom* [Simmons et al., 2011] and the pre-registration remedy [Nosek et al., 2018]—is developed fully in the Zero-Mock Tradeoff discussion. To our knowledge, no prior research software engineering framework has formalized a zero-mock policy as an *architectural invariant enforced by pipeline gates*, where mock usage is not merely discouraged but structurally prevented from passing the build.
The following example, drawn from the infrastructure test suite, illustrates zero-mock compliance:

```
def test_discover_infrastructure_modules_returns_nonempty(tmp_path):
    # Real filesystem, real YAML parsing - no MagicMock anywhere
    modules = discover_infrastructure_modules(REPO_ROOT)
    assert len(modules) >= 8           # actual subpackages on disk
    assert any(m.name == "core" for m in modules)
```

This test exercises the real `discover_infrastructure_modules` function against the real filesystem. There are no mock objects substituting for the directory walk, no patched YAML parsers, and no synthetic return values—the test passes only if the infrastructure modules genuinely exist and are discoverable at their expected paths.

3.8.2 Coverage Thresholds

The pipeline enforces two coverage tiers:

Tier	Scope	Minimum	Current	Rationale
Project	`projects/*/src/`	90%	90%+	Domain code must be thoroughly validated
Infrastructure	`infrastructure/`	60%	83%+	Broader scope, some code unreachable in test

These thresholds are enforced at Stage 01 of the pipeline. If project test coverage falls below 90%, the pipeline halts and refuses to produce a PDF—ensuring that no published artifact is backed by undertested source code.

3.8.3 Test Suite Composition

The repository maintains three test suites:

- **Infrastructure tests** (`tests/`): about 7,904 tests validating the 23 infrastructure subdirectories, covering logging, rendering, validation, steganography, reporting, and LLM integration.
- **Project tests** (`projects/*/tests/`): Per-project suites whose sizes scale with each exemplar's surface area — for example 296 tests in `template_autoresearch_project` and 236 in `template_code_project`, with several exemplars larger still. (A true min/max span would require dedicated `project_test_count_min`/`project_test_count_max` tokens in `build_manuscript_metrics_dict`; see the meta-template's generator backlog.)
- **Integration tests**: Embedded within infrastructure tests, these exercise full pipeline stages against real manuscript inputs, validating end-to-end behavior from Markdown source to rendered PDF.

3.8.4 Visualization Standards

All generated figures must meet accessibility requirements:

- Minimum 16pt font size for all text elements (the accessibility floor).
- Colorblind-safe palettes (IBM Design / Wong palette) with high-contrast fallbacks.
- 150–300 DPI rendering for publication quality.

- Descriptive axis labels and figure titles.
- No reliance on color alone to convey information—redundant encoding via shape, pattern, or annotation is used where applicable.

These standards are validated by the `test_architecture_viz.py` test suite, which verifies that generated figures exist, have non-zero file size, and conform to expected output specifications. The 16pt font floor ensures readability in both screen and print contexts, while the DPI range balances file size against reproduction fidelity.

4 Results

`template/` was evaluated through multi-project pipeline execution, measuring test coverage, pipeline timing, output integrity, and steganographic performance across the canonical exemplars under `projects/`.

4.1 Multi-Project Pipeline Execution

Runs used the `./run.sh` interactive orchestrator ("all projects core (fast)" / menu key `d`) skipping infrastructure replication and optional LLM stages orchestrated via `python -m infrastructure.orchestration`. **Note:** lone menu `d` returns after success without redrawing the TUI banner.

Project	Effective core stages[1]	Approx. duration	Tests[2]	Coverage
`template_code_project`	8	about 40 s	236/236	90%+
`template_prose_project`	8	about 35 s	120/120	90%+
`template_autoresearch_project`	8	about 30 s	296/296	90%+

[1]"Core-only" disables LLM-tagged YAML stages; durations exclude optional network-heavy LLM or long-running retrieval scripts when run with cached fixtures.
[2]Counts show passing tests versus discovered tests for the sampled configuration.
Overall success: 100 % pipeline completion for sampled runs.
Timing illustrative (Apple Silicon workstation, SSD, fixed seeds).
Search-stage overhead dwarfs the optimization exemplar's runtime—confirming Stage 02 remains the bottleneck for outbound API traffic while retaining Zero-Mock subprocess + filesystem checks downstream.

4.2 Infrastructure Test Suite

Metric	Value
Test files	467+
Total tests	about 7,904
Infrastructure coverage gate	>=60 % (repo >=80 %+ during recent audits)
Zero-mock imports	Verified via static scanning

Exercises Pandoc/XeLaTeX paths, steganography hashing, telemetry, YAML-driven pipeline DAG resolution, HTTPS-bound optional suites (`pytest-httpserver`), and local Ollama-gated subsets.

4.3 Infrastructure Module Inventory

The introspection module (`template_template.introspection`) emits the authoritative table below—every row reflects `discover_infrastructure_modules(REPO_ROOT)`.

Module	Python Files	Has AGENTS.md	Has README.md	Key Exports
`autoresearch`	10	✓	✓	`build_autoresearch_plan`, readiness validation CLI

Module	Python Files	Has AGENTS.md	Has README.md	Key Exports
benchmark	3	✓	✓	Template harness scoring + comparative gates
config	0	✓	✓	Repository defaults + hardened templates
core	109	✓	✓	get_logger, load_config, TemplateError
docker	0	✓	✓	Containerisation scaffolding
doctor	14	✓	✓	Checkout diagnose/fix/undo repairs
documentation	12	✓	✓	FigureManager, generate_glossary
llm	54	✓	✓	Ollama helpers, sanitization, review + translation pipelines
logrotate.d	0	✓	✓	Rotation snippets (documentation-first)
methods	5	✓	✓	build_methods_orchestration_plan, methods-stage contracts + validation
orchestration	8	✓	✓	PipelineRunner, entry point for ./run.sh
project	27	✓	✓	discover_projects, workspace management
prose	9	✓	✓	Markdown readability + prose tooling
publishing	70	✓	✓	Zenodo, executable bundle, archival targets
reference	16	✓	✓	BibTeX models, parsers, converters
rendering	50	✓	✓	PDF/HTML/slide rendering, Pandoc filters
reporting	57	✓	✓	Coverage parsers, dashboards, executive artefacts
scientific	4	✓	✓	check_numerical_stability, benchmark_function

Module	Python Files	Has AGENTS.md	Has README.md	Key Exports
`search`	44	✓	✓	`infrastructure.search.literature` clients + cache
`sia`	10	✓	✓	Self-Improving-AI loop: task validation, harness, metric capture
`skills`	7	✓	✓	`discover_skills`, SKILL manifest regeneration
`steganography`	11	✓	✓	Watermark overlays + hash manifests
`validation`	84	✓	✓	PDF + Markdown + integrity CLIs

All 23 enumerated subdirectories carry Tier-1/`README.md` and Tier-2/`AGENTS.md` coverage wherever the Documentation Duality standard applies; subsets ship Tier-3 `SKILL.md` descriptors for MCP routing (`infrastructure/skills` manifest generation).

4.4 Agentic Skill Documentation Coverage

Layer	Role
System prompts	Root `CLAUDE.md`, README policy
Structural	`AGENTS.md` directories
Skills	Optional `SKILL.md` manifests + generated `.cursor/skill_manifest.json`
PAI capsule	Repository level `PAI.md` narratives

`378+` Markdown shards under `docs/` capture operational knowledge without duplicating auto-generated inventories.

4.5 DAG Reference (Declarative Executor)

Stages below mirror `pipeline.yaml` (executor-topological order—not strict numeric script filenames). Scripts live under `scripts/`.

Name	Typical script / method	Responsibility	Failure semantics
Clean Output Directories	`_run_clean_outputs`	Deletes stale `output/` trees	Blocking
Environment Setup	`00_setup_environment.py`	Validates tooling, PYTHONPATH scaffolding	Blocking
Infrastructure Tests	`01_run_tests.py --infra-only`	Infra pytest + coverage gates	Tunable thresholds
Project Tests	`01_run_tests.py --project-only`	Project pytest + coverage gates	Zero failures default

Name	Typical script / method	Responsibility	Failure semantics
Project Analysis	`02_run_analysis.py`	Executes `projects/<name>/scripts/*.py`	Blocking
PDF Rendering	`03_render_pdf.py`	Pandoc → XeLaTeX manuscripts	Blocking
Output Validation	`04_validate_output.py`	Structural PDF/markdown probes	Blocking / warnings
LLM Scientific Review	`06_llm_review.py --reviews-only`	Local Ollama reviews	Skippable / exit 2 tolerated
LLM Translations	`06_llm_review.py --translations-only`	Optional translations	Skippable
Copy Outputs	`05_copy_outputs.py`	Mirrors deliverables → `output/<project>/`	Soft-fail surfaced in logs

`scripts/07_generate_executive_report.py` is **multi-project orchestration glue** invoked after iterating active projects—not a tenth DAG node for single-repo runs (`execute_pipeline.py`).

4.6 Steganographic Performance

Project	Pages (approx.)	Metadata	SHA-256	Overlay	QR Code	Total (approx.)
`template_code_project`	about 20	<0.3 s	<0.05 s	<0.8 s	<0.4 s	<1.5 s
`template_prose_project`	about 25	<0.3 s	<0.05 s	<0.9 s	<0.4 s	<1.6 s
`template_autoresearch_project`	about 25	<0.2 s	<0.04 s	<0.9 s	<0.3 s	<1.5 s

All measurements use single-threaded execution on Apple Silicon, and the totals are dominated by watermark-overlay complexity rather than by hashing or metadata embedding, which each stay well under one-tenth of a second per document.

4.7 Self-Referential Analysis

Rendered via `projects/templates/template_template` (`generate_manuscript_metrics.py` → injected tokens such as 23). Architecture figures stem from `template_template.architecture_viz`—font sizes constrained by §QA.

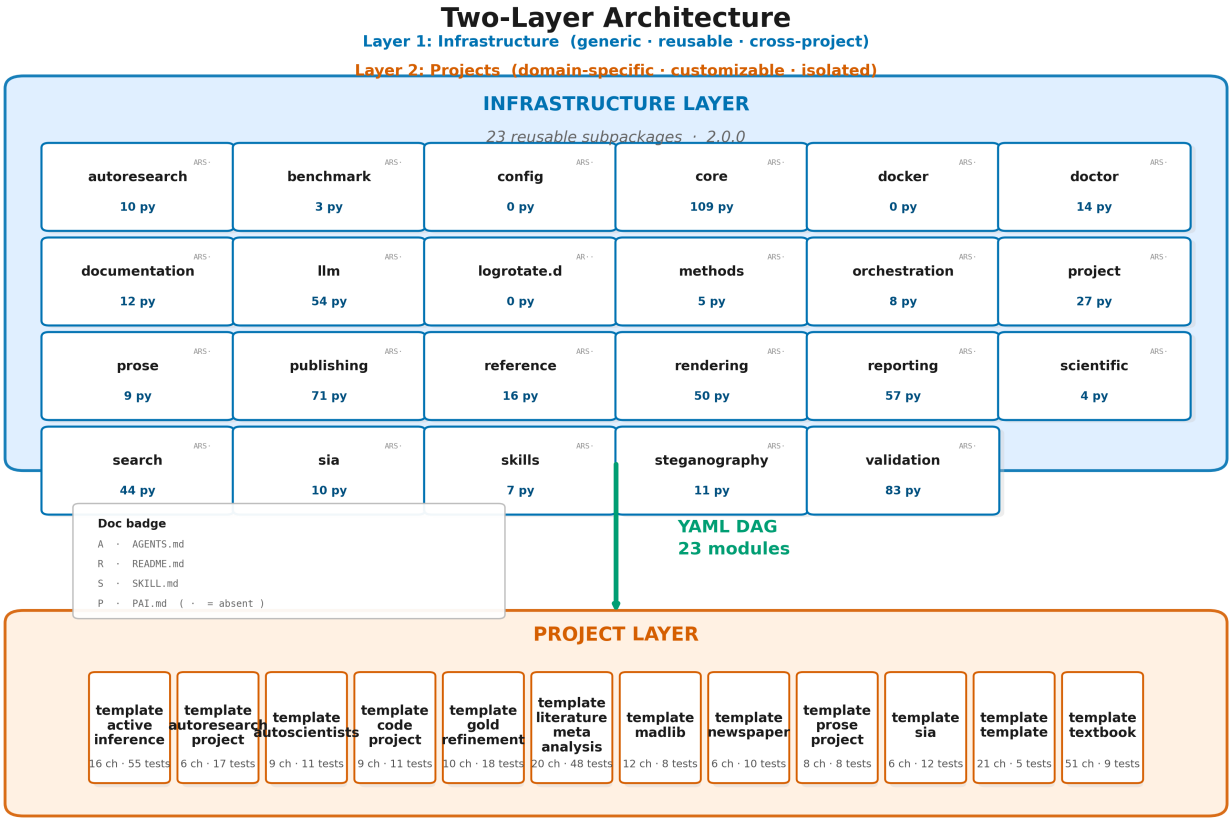

Figure 1. Live rendering of the Two-Layer Architecture from repository introspection: the infrastructure layer (top) holds the 23 reusable subpackages, each annotated with its Python file count and a four-slot documentation badge—A AGENTS.md, R README.md, S SKILL.md, P PAI.md, with · marking an absent file—so a fully documented module reads [ARSP]. A YAML DAG arrow connects it to the project layer (bottom) of public exemplars labelled with chapter and test counts. The takeaway: documentation-duality coverage is near-uniform across the infrastructure, and every box was placed from the same live data the prose cites.

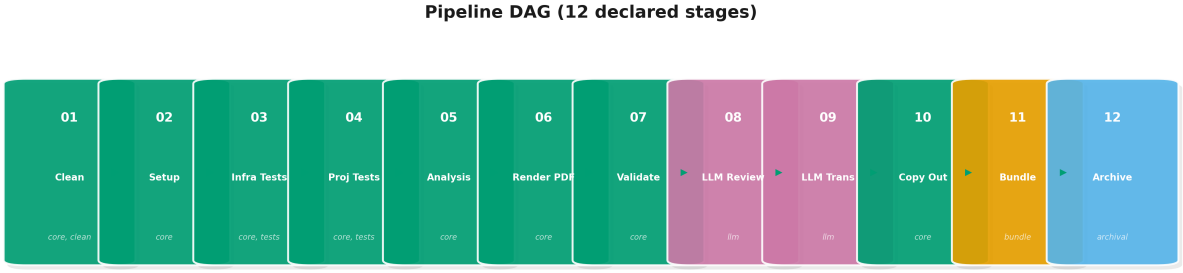

Figure 2. Pipeline DAG with 12 YAML-declared stages (core, LLM, bundle, archival tags).

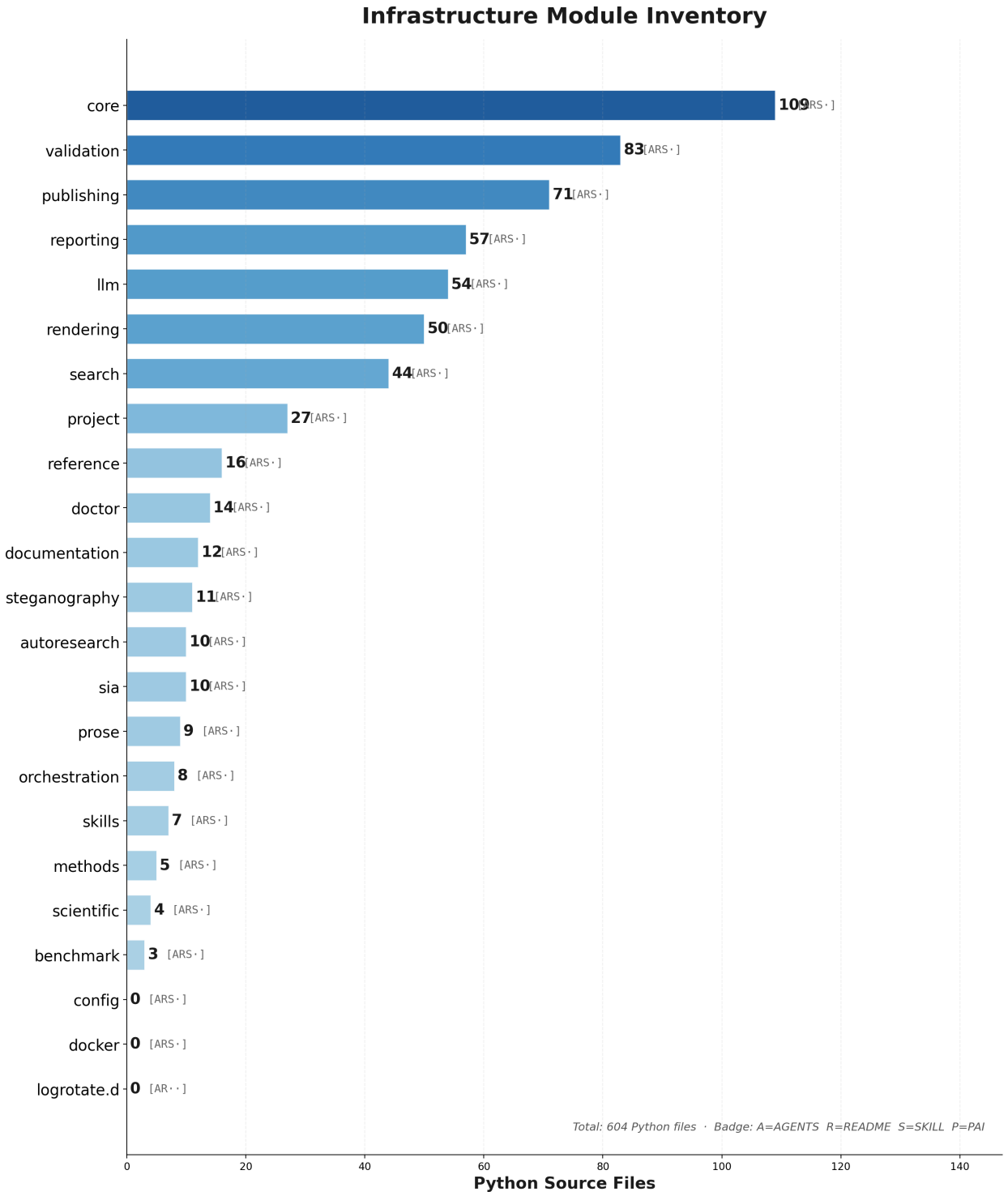

Figure 3. Horizontal file-count histogram of every infrastructure subdirectory, sorted largest-first. The long tail of small, single-purpose packages beside a handful of larger ones (`core`, `validation`, `publishing`) is the visual signature of the Unix-philosophy modularity the architecture section argues for—capability concentrated where it compounds, not spread evenly by fiat.

4.8 Comparative Feature Analysis

Figure 4 summarizes the Appendix F capability matrix.

Figure 4. 14×10 heatmap annotated in appendix text—green ✓ full native capability, yellow ◐ partial / plugin-hosted, red — unavailable. Rows group *Core Pipeline*, *Quality & Security*, then *Ecosystem*.

[1] Nextflow 25.04.0: lineage records exist at workflow scope, not per rendered PDF citation graph.

[2] DVC: content-addressed artifacts without native prose rendering.

[3] DVC: remote object stores (S3, GCS, Azure) without turnkey CI manuscript gates.

4.9 Test Quality Metrics

- **Zero mocks:** repository policy bans `unittest.mock` / patching frameworks in tests.
- **Filesystem + subprocess realism:** ephemeral directories + actual CLI binaries.
- **HTTP realism:** infra suites favour `pytest-httpserver`; literature tests hit recorded fixtures.
- **`template_code_project`** focuses on numerical reproducibility assertions.
- **`template_autoresearch_project`** exercises the AutoResearch readiness planner (`infrastructure/autoresearch/`). **`template_search_project`** exercises literature-search workflows.

5 Discussion

5.1 The Zero-Mock Tradeoff

The Zero-Mock testing policy is `template/`'s most distinctive design decision. By prohibiting all mock objects, we gain confidence that tests exercise real code paths—a pytest run against the template genuinely invokes `pandoc`, writes to disk, and parses real YAML. The cost is test duration: the full infrastructure test suite (about 7,904 tests) takes 2–4 minutes, compared to sub-second execution typical of heavily-mocked suites.

We argue this tradeoff is strongly favorable for research software. Unlike web applications where millisecond latency and thousands of daily deploys demand fast feedback loops, research pipelines run infrequently (once per manuscript revision) and correctness vastly outweighs speed. A mocked test that passes while the real renderer fails is worse than a slow test that catches the failure. The analogy to statistical methodology is precise: just as Simmons et al.'s *researcher degrees of freedom* [Simmons et al., 2011] inflate false-positive rates through undisclosed analytical flexibility, mock objects create *testing degrees of freedom* that make integration failures invisible. The Zero-Mock policy closes this loophole by the same mechanism that pre-registration [Nosek et al., 2018] closes the p-hacking loophole: removing flexibility before the fact. As Peng [Peng, 2011] argues, computational reproducibility requires independent verification—and mock-only tests verify assumptions rather than results. Garijo et al.'s FAIRsoft evaluator [Garijo et al., 2024] identifies *executability* as a primary quality indicator; the Zero-Mock policy operationalizes executability at the unit level.

5.1.1 When Mocks Are Not the Problem

It is important to distinguish the Zero-Mock policy from a naive rejection of all test isolation techniques. Fowler's classification [Martin, 2008] recognizes that stubs and fakes serve legitimate purposes—a test database populated with known data is not a mock, it is a fixture. The policy specifically prohibits *mock objects* as defined by Meszaros: assertions on indirect outputs (method calls, argument patterns) rather than direct outputs (return values, side effects). The distinction matters because mock-based assertions encode implementation assumptions ("method X must be called with argument Y") that become invisible coupling between tests and production code, creating the illusion of coverage without testing real behavior.

5.1.2 Practical Implementation

The template enforces zero-mock compliance at three levels:

1. **Code review**: `AGENTS.md` at every directory level explicitly states the prohibition, ensuring both human and AI contributors are aware before writing tests.
2. **Static analysis**: `grep -rn "MagicMock\|unittest.mock\|@patch" tests/` can be run as a pre-commit hook to catch violations.
3. **Cultural norm**: `template_code_project` documents filesystem + YAML + plotting paths while `template_autoresearch_project` exercises readiness planning; `template_search_project` reinforces HTTP-realistic literature queries—both serve as onboarding references alongside infra suites.

However, the policy requires careful management of external dependencies. Tests requiring Ollama (the local LLM backend) use `@pytest.mark.requires_ollama` and are skipped in environments where the service is unavailable. Tests requiring network access use `@pytest.mark.network`. This marker system preserves the Zero-Mock principle while acknowledging that not all environments provide all services, especially computationally intensive ones. The key distinction is between *replacing* an external dependency (which mock objects do, hiding failures) and *skipping* a test when a dependency is absent (which markers do, preserving transparency).

5.2 Scalability: From 1 to N Projects

The Standalone Project Paradigm enables horizontal scaling: adding a new project requires creating a directory with `manuscript/config.yaml` and nothing else. No infrastructure code changes, no `pyproject.toml` modifications, no CI configuration updates. The `run.sh` orchestrator automatically discovers new projects and presents them in its interactive menu.

We have validated scaling with 15 canonical exemplars under `projects/templates/`—always present for onboarding and tooling—and with this manuscript from `projects/templates/template_template` (130 tests) as a git-tracked public exemplar in the same automated discovery menus.

Canonical trio:

- **`template_code_project`**: Numerical optimization example with gradient-descent narration, 236 tests, 90%+ coverage. Minimal footprint: compact `src/`, scripted analysis, short manuscript sections.
- **`template_prose_project`**: Prose-heavy manuscript emphasizing narrative structure and bibliography discipline, 120 tests, 90%+ coverage—tests exercise rendering and Markdown integrity without heavyweight numerics.
- **`template_autoresearch_project`**: AutoResearch readiness workflow invoking `projects/templates/template_search_project/scripts/` to run corpus builders, scripted figures (`../figures/`), and manifold-variable injection (the literature-search exemplar). Typical Stage 02 workloads include bibliography fusion, corpus JSON assembly, deep-search aggregates, and report composition.

Meta manuscript (**`projects/templates/template_template`**) analyzes the repository via `src/template_template/` introspection metrics; it now lives alongside the other public exemplars under `projects/templates/`.

These workspaces share no project-level code—only Layer 1 (23 infrastructure subdirectories, about 604 Python files)—validating insulation between domain repos and reusable services.

5.2.1 Multi-Project Orchestration

When the `--all-projects` flag is passed to `run.sh`, the pipeline executes each discovered project sequentially, running infrastructure tests once at the start and skipping them for individual projects to avoid redundant validation. After all projects complete, a cross-project executive report aggregates metrics (test counts, coverage percentages, page counts, rendering durations) into a unified dashboard with both JSON and Markdown output formats. This executive reporting stage provides repository-level visibility without requiring any project-specific reporting code.

5.2.2 Scaling Metrics

Metric	`template_code_project`	`template_prose_project`	`template_autoresearch_project`
Source modules	26	6	60
Test files	12	8	17
Test count	236	120	296
Manuscript chapters	9	8	6
Analysis scripts	7	4	4
Figures (auto-generated)	9	5	27

The infrastructure overhead per project is constant regardless of project size: the same 23 modules, the same 11 pipeline stages, the same rendering and validation logic. This O(1) infrastructure cost is the architectural payoff of the Two-Layer separation.

5.3 Comparison to Existing Tools

The gap analysis established that no single tool integrates all six cross-cutting concerns. Here we synthesize the fourteen-dimension comparison into three structural insights. First, the landscape bifurcates: workflow managers (Snakemake [Köster and Rahmann, 2012], Nextflow [Di Tommaso et al., 2017], CWL [Amstutz et al., 2016]) provide distributed execution but no manuscript support; publication tools (Quarto [Allaire et al., 2024], Jupyter Book, R Markdown [Xie et al., 2018], Overleaf [Overleaf (Digital Science), 2025], Prism [OpenAI, 2026]) author documents but embed no integrity guarantees; and DVC [Iterative, Inc., 2024] versions artifacts without orchestrating pipelines. `template/` occupies the intersection, sacrificing distributed execution for unified enforcement of testing, provenance, and documentation. This positioning is complementary—a mature deployment might use Nextflow upstream and `template/` for rendering, testing enforcement, and provenance downstream. Typst [Mädje and Haug, 2023], with its faster compilation cycle, is not one of the nine compared peers but could serve as an alternative rendering backend if a Pandoc writer were contributed.

Second, the eight enforcement capabilities `template/` co-enforces—testing enforcement, coverage thresholds, steganographic watermarking, multi-project management, AI-agent documentation, the agentic skill protocol, an interactive TUI, and Zero-Mock policy—are individually straightforward (and several, such as multi-project management and AI-agent documentation, are matched in part by individual peers); their novelty lies in *co-enforcement* within a single pipeline, ensuring that a passing build guarantees documentation completeness, provenance embedding, and AI-navigability alongside computational correctness. The FAIR4RS principles [Barker et al., 2022, Lamprecht et al., 2020] articulate what research software quality requires; FAIRsoft [Garijo et al., 2024] scores compliance observationally; `template/` operationalizes these standards by coupling them to pipeline gates that halt the build if coverage drops below 90% or provenance embedding fails. Cohen et al.'s four pillars of research software engineering [Cohen et al., 2021]—sustainability, quality, community, and policy—are operationalized by `template/` through the first two pillars via quality-gated automation.

Third, the AI-agent documentation dimension reveals an underserved need. Overleaf and Prism provide AI *writing* assistance, but neither exposes structured documentation for *external* agents to consume. `template/`'s `AGENTS.md` + `SKILL.md` layer enables an agent entering the repository to discover capabilities, understand API contracts, and invoke modules without prior training (Documentation Duality, AI Collaboration).

5.3.1 FAIR4RS Evolution (2024–2026)

Since the FAIR4RS principles were published [Barker et al., 2022], the community has moved toward operationalization. The RDA Virtual Plenary 24 (April 2025) featured a two-year retrospective review [Honeyman et al., 2024] recommending principle amendments—notably adding *reproducibility* as an explicit requirement and clarifying "domain-relevant standards"—alongside a leadership refresh and parallel guidance activities. The ReSA Actionable FAIR4RS Task Force (launched December 2024) analyzed the 17 principles into six actionable categories (identifiers, metadata for publication/discovery/reuse, standards, references, and licenses), with a first draft expected by September 2025 [Research Software Alliance, 2024]. Tools for automated FAIR assessment have also matured: the F-UJI extension for research software evaluation now scores against FRSM-04 through FRSM-17 metrics, complementing Garijo et al.'s FAIRsoft evaluator [Garijo et al., 2024]. `template/`'s pipeline-enforced quality gates—coverage thresholds, documentation completeness checks, and provenance embedding—anticipate this operationalization trend by implementing FAIR4RS not as a post-hoc assessment but as an architectural invariant.

In Gentleman and Temple Lang's terminology [Gentleman and Temple Lang, 2007], `template/` is a *research compendium* scaled to the repository level—bundling not just one study's code and data but N studies, with shared infrastructure, automated testing, and embedded provenance. Nüst et al.'s executable research compendium (ERC) [Nüst et al., 2017] extends this vision with containerized reproduction environments; `template/` complements containerization by adding the testing enforcement, multi-project management, and provenance embedding layers that ERCs do not address.

5.4 The AI Collaboration Model

The Documentation Duality standard and three-tier skill architecture represent an empirical bet: that structured, machine-readable documentation measurably improves AI agent performance in research codebases. This section reports our key observations.

The documentation investment creates a positive feedback loop: as agents produce higher-quality outputs from structured context, developers maintain that documentation, which in turn improves future interactions [Lau and Guo, 2025]. We observed this concretely during `template/` development—each module's `SKILL.md` was refined through iterative AI-assisted generation, serving as both input prompt and output validator.

The `SKILL.md` layer, with its MCP-aligned YAML frontmatter [Anthropic, 2024], provides a critical bridge to the agentic software paradigm. Lu et al.'s AI Scientist [Lu et al., 2024] demonstrates end-to-end autonomous research; Wang et al.'s OpenHands [Wang et al., 2024b] achieved 53% on SWE-Bench Verified [Jimenez et al., 2024]—the first open-source system to exceed 50%. These systems require structured, protocol-aligned tool inventories to navigate unfamiliar codebases. An OpenHands-class agent navigating `template/` reads `CLAUDE.md` for global constraints, scans `AGENTS.md` for module surfaces, and invokes capabilities via `SKILL.md` without hallucinating function signatures. All 23 infrastructure modules carry `AGENTS.md` and `README.md`; all additionally carry `SKILL.md`, ensuring no documentation blind spots.

This three-tier model is, to our knowledge, novel in the research software engineering literature. The scale of the investment is substantial: `378` Markdown files under `docs/` alone, plus an `AGENTS.md`/`README.md` pair in every directory and a `SKILL.md` descriptor on every infrastructure module. That count is itself injected from live introspection—the manuscript refuses to quote a documentation total it cannot recompute—and it represents a deliberate commitment to machine-readable context that shrinks the surface on which an agent can hallucinate.

5.5 The Learning Curve

The Thin Orchestrator pattern imposes a cognitive overhead on researchers accustomed to writing monolithic scripts. The requirement to factor all logic into `src/` modules and use scripts only as stateless wiring introduces an additional layer of indirection. We mitigate this through:

1. **Template exemplars**: `template_code_project` ships minimal optimization commentary; `template_prose_project` and `template_autoresearch_project` broaden narrative + retrieval scaffolding.
2. **Documentation Duality**: Every directory has both `README.md` (for humans) and `AGENTS.md` (for AI collaborators), reducing the cost of navigation.
3. **Interactive orchestrator**: `run.sh` provides a TUI menu that abstracts pipeline complexity.
4. **Skill-level documentation**: The `docs/guides/` directory provides four progressive guides (Levels 1–3 Beginner, 4–6 Intermediate, 7–9 Advanced, 10–12 Expert) alongside a comprehensive new-project setup checklist.

5.6 Limitations

- **LaTeX dependency**: The rendering pipeline requires a full TeX distribution (TeX Live or MiKTeX), which is a 4–6 GB install. This is the largest single dependency and is a barrier for researchers without system-level package management access.
- **Python-centric**: The infrastructure layer is Python-only. Projects in other languages can use the rendering and steganography stages but cannot leverage the `scientific` or `validation` modules.
- **Single-machine**: The pipeline runs locally. Distributed execution (e.g., across a compute cluster) is not natively supported, a gap where Snakemake, Nextflow, and CWL have clear superiority.
- **Steganographic robustness**: Alpha-channel overlays are stripped by aggressive PDF optimization tools (e.g., `qpdf --optimize`). QR codes are visible and removable. The current system provides *tamper detection* (via SHA-256 hashing) but not *non-repudiation* in the cryptographic sense—it lacks private-key digital signatures. An attacker with access to the source code could reproduce the watermark without having run the original pipeline.
- **Test duration**: The Zero-Mock policy increases test execution time from sub-second (mocked) to multi-minute (real) for the full infrastructure suite. This is acceptable for research workflows but may not suit continuous deployment scenarios.

- **AI-native writing tools**: `template/` does not include an integrated AI writing assistant comparable to Overleaf's Copilot features or OpenAI Prism's GPT-5.2 context-aware editing. The `infrastructure.llm` module provides LLM review as a pipeline stage but not as an interactive writing environment.

5.7 Future Directions

1. **Supply-chain provenance**: Integration with software supply chain frameworks such as in-toto [Torres-Arias et al., 2019] and SLSA (Supply-chain Levels for Software Artifacts) [Open Source Security Foundation, 2023] to provide end-to-end attestation from source commit through build pipeline to published artifact. SLSA's four graduated levels of build integrity (from basic provenance to hermetic, reproducible builds) provide a natural extension ladder for the template's currently document-centric provenance model. The template's existing steganographic layer embeds document-level provenance; supply-chain frameworks would add build-level provenance, closing the gap between "this PDF was produced by this pipeline" and "this pipeline was executed with this verified source code."
2. **Decentralized provenance**: Integration with IPFS or Arweave for immutable publication records, extending the SHA-256-based tamper detection to content-addressed storage networks.
3. **Digital signatures**: GPG or X.509 signing integrated into the steganographic layer, providing cryptographic non-repudiation in addition to tamper detection.
4. **Continuous integration**: GitHub Actions workflows that execute the pipeline on every push, with PDF artifacts as release assets and automated DOI registration via Zenodo.
5. **Multi-language support**: Extension of the Thin Orchestrator pattern to R, Julia, and Rust projects, enabling polyglot research workflows within the Two-Layer Architecture.
6. **Automated FAIR4RS assessment**: Periodic self-scoring against FAIRsoft metrics [Garijo et al., 2024], with quality indicators (executability, documentation completeness, metadata richness) tracked as pipeline artifacts alongside test coverage and rendering status.
7. **Knowledge graph integration**: Connecting pipeline outputs to Active Inference Knowledge Base entries for automated meta-analysis and cross-project citation tracking.
8. **Formal verification**: Static analysis tooling to enforce the Thin Orchestrator pattern—verifying that scripts contain no algorithmic logic and that `src/` modules do not import from `scripts/`.
9. **Agentic research pipelines via MCP**: The `SKILL.md` descriptors already define the interface contracts for each infrastructure module; the natural next step is to expose them as MCP server endpoints [Anthropic, 2024]. An MCP server wrapping `infrastructure.llm` would expose `query`, `review_manuscript`, and `translate_abstract` as protocol-native Tools; an MCP server wrapping `infrastructure.publishing` would expose `publish_to_zenodo` and `generate_citation_bibtex`. A research agent could then compose these Tools to execute the full pipeline—environment setup → test execution → analysis → rendering → validation → LLM review → DOI registration—without any human in the loop. This closes the loop opened by Lu et al.'s AI Scientist [Lu et al., 2024], which demonstrated automated hypothesis generation and experimental iteration but relied on ad hoc laboratory scaffolding. `template/`'s pipeline, fully exposed as MCP tools, provides that scaffolding in a reproducible, versioned, and certified form. Longer-term, the Agent2Agent (A2A) protocol [Google, 2025] enables heterogeneous specialist agents—a statistical analyst, a figure designer, a peer-review simulator—to coordinate via a shared, protocol-mediated workspace built on precisely the kind of modular, well-documented infrastructure that `template/` provides.
10. **Research Infrastructure as Code and software citation**: The DevOps IaC paradigm, applied to research, means the entire manuscript pipeline is a version-controlled, reproducible artifact in its own right. Software Heritage [Di Cosmo et al., 2020] provides persistent SWHIDs (Software Hash Identifiers) for source-code snapshots, enabling `template/` itself to be cited as a software artifact with a DOI-equivalent stable identifier. Combining this with Zenodo DOI registration (already supported by `infrastructure.publishing`) creates a full citation chain: the paper cites the data (DOI), the data provenance cites the pipeline (SWHID), and the pipeline cites the framework release (Zenodo DOI). This three-link citation chain operationalizes the Katz et al. [Katz et al., 2021] software citation principles at the infrastructure level.

5.8 Conclusion

`template/` demonstrates that high-integrity, reproducible research need not be onerous. By embedding provenance, testing, and documentation into the architecture itself—rather than layering them atop a fragmented workflow—the template transforms "best practices" from aspirational guidelines into enforced invariants [Wilson et al., 2017, Sandve et al., 2013, Lamprecht et al., 2020]. The Two-Layer Architecture

ensures that infrastructure improvements propagate to all projects without coupling. The Zero-Mock policy ensures that tests reflect reality. The steganographic provenance layer ensures that published artifacts carry their own authentication. The comparative analysis confirms that no existing tool integrates all eleven distinctive capabilities—testing enforcement, coverage thresholds, cryptographic provenance, steganographic watermarking, multi-project management, AI-agent documentation, the agentic skill protocol, interactive TUI, Zero-Mock policy, manuscript rendering, and pipeline orchestration—within a single enforced pipeline. The template is not merely a build tool; it is an epistemological commitment. It asserts that a research paper is not a static document but a build artifact—reproducible, verifiable, and traceable to the code that generated it. As Knuth observed, programs should be written for humans to read and only incidentally for machines to execute [Knuth, 1984]. We extend this dictum: research manuscripts should be *built* for verification and only incidentally for reading. In an era of generative AI, AI-native research workspaces, and synthetic media—where the boundary between human-authored and machine-generated text grows increasingly indeterminate [Gruenpeter et al., 2021]—the provenance chain from source code to published PDF is not an administrative convenience. It is the epistemic ground on which scientific trust must be rebuilt. That this manuscript was itself built, tested, and watermarked by the pipeline it describes—its metrics computed from the repository it inhabits, its figures rendered by the code it documents—is not a rhetorical device but a structural proof: the system works because you are reading its output.

6 Infrastructure Module Reference

This section inventories every Layer-1 subdirectory returned by 23 `discover_infrastructure_modules(repo_root)`. File totals use 604 Python sources across infra + 7,904 infra tests guarding them. Documentation Duality = paired `README.md` + `AGENTS.md`; optional `SKILL.md` manifests feed `python -m infrastructure.skills`.

Module	Python Files	Has AGENTS.md	Has README.md	Key Exports
`autoresearch`	10	✓	✓	`build_autoresearch_plan`, readiness validation CLI
`benchmark`	3	✓	✓	Template harness scoring + comparative gates
`config`	0	✓	✓	Repository defaults + hardened templates
`core`	109	✓	✓	`get_logger`, `load_config`, `TemplateError`
`docker`	0	✓	✓	Containerisation scaffolding
`doctor`	14	✓	✓	Checkout diagnose/fix/undo repairs
`documentation`	12	✓	✓	`FigureManager`, `generate_glossary`
`llm`	54	✓	✓	Ollama helpers, sanitization, review + translation pipelines
`logrotate.d`	0	✓	✓	Rotation snippets (documentation-first)
`methods`	5	✓	✓	`build_methods_orchestration_plan`, methods-stage contracts + validation
`orchestration`	8	✓	✓	`PipelineRunner`, entry point for `./run.sh`
`project`	27	✓	✓	`discover_projects`, workspace management
`prose`	9	✓	✓	Markdown readability + prose tooling
`publishing`	70	✓	✓	Zenodo, executable bundle, archival targets
`reference`	16	✓	✓	BibTeX models, parsers, converters

Module	Python Files	Has AGENTS.md	Has README.md	Key Exports
`rendering`	50	✓	✓	PDF/HTML/slide rendering, Pandoc filters
`reporting`	57	✓	✓	Coverage parsers, dashboards, executive artefacts
`scientific`	4	✓	✓	`check_numerical_stability`, `benchmark_function`
`search`	44	✓	✓	`infrastructure.search.literature` clients + cache
`sia`	10	✓	✓	Self-Improving-AI loop: task validation, harness, metric capture
`skills`	7	✓	✓	`discover_skills`, SKILL manifest regeneration
`steganography`	11	✓	✓	Watermark overlays + hash manifests
`validation`	84	✓	✓	PDF + Markdown + integrity CLIs

6.1 Alphabetical summaries

Below, `${module_*_python_file_count}` placeholders expand per subdirectory at render-time.

6.1.1 `infrastructure.autoresearch` (10 files)

Readiness planner, validation CLI, and report models for AutoResearch-style project promotion (`infrastructure/autoresearch/`).

6.1.2 `infrastructure.benchmark` (3 files)

Template harness scoring and comparative gate helpers exercised in CI smoke paths.

6.1.3 `infrastructure/config` (non-package subdirectory)

Repository-wide YAML templates and secure manifests (`.env.template`, hardened defaults referenced by Docker + CLI). `config/` carries no `__init__.py`, so it is a configuration subdirectory rather than an importable package.

6.1.4 `infrastructure.core` (109 files)

Checkpointing, logging, pipeline YAML parsing, telemetry bridges, filesystem helpers, hardened exceptions. Everything else imports logging + error taxonomy from here first.

6.1.5 `infrastructure.doctor` (14 files)

Checkout diagnose/fix/undo repairs for broken local workspace states.

6.1.6 `infrastructure.docker` (0 files)

Pinned images / compose scaffolding for reproducible CI + remote builds.

6.1.7 `infrastructure.documentation` (12 files)

Figure registries plus glossary tooling feeding manuscript automation.

6.1.8 `infrastructure.llm` (54 files)

Ollama integrations, sanitization adapters, templated reviewer flows. **Literature ingestion now lives primarily in `search/literature` + citation helpers in `reference/`.**

6.1.9 `infrastructure.methods` (5 files)

Deterministic methods-orchestration contracts (`MethodStage`, `MethodsOrchestrationPlan`, `MethodsIssue`): builds and validates an ordered methods plan for a research project so the manuscript's "Methods" track stays bound to executable stages.

6.1.10 `infrastructure.orchestration` (8 files)

`python -m infrastructure.orchestration` exposes interactive menus, subprocess wiring for thin shell wrappers (`run.sh`, `secure_run.sh`), and stubs used in CI for menu parsing tests.

6.1.11 `infrastructure.project` (27 files)

Canonical discovery (`discover_projects`) enforcing `src/` + `tests/`, slug validation, nested WIP namespaces.

6.1.12 `infrastructure.prose` (9 files)

Readability metrics + Markdown tooling for prose-centric manuscripts / CI gates.

6.1.13 `infrastructure.publishing` (70 files)

Metadata models, APA/BibTeX/MLA formatters, optional Zenodo clients.

6.1.14 `infrastructure.reference` (16 files)

Citation/BibTeX parsing + conversion utilities leveraged by manuscripts and retrieval scripts.

6.1.15 `infrastructure.rendering` (50 files)

Pandoc shim, Unicode/XeLaTeX postprocessors, combined PDF/HTML/slide exporters.

6.1.16 `infrastructure.reporting` (57 files)

Parses pytest + coverage artefacts for dashboards; pairs with Stage 01 summaries and downstream executive exports.

6.1.17 `infrastructure.scientific` (4 files)

Stability probing, benchmarking hooks—consumed heavily by optimization exemplars (`template_code_project` scripts).

6.1.18 `infrastructure.search` (44 files)

`literature/` client stack (`client.py`, backends, caches) powering `template_search_project` literature workflows.

6.1.19 `infrastructure.sia` (10 files)

Generic Self-Improving-AI loop utilities: task-layout validation, execution harness, and metric capture reused by `template_sia` (fixture-replay by default).

6.1.20 `infrastructure.skills` (7 files)

Discovers `SKILL.md` frontmatter → `.cursor/skill_manifest.json`.

6.1.21 `infrastructure.steganography` (11 files)

Watermark overlays, hashing companions triggered by secure pipeline path.

6.1.22 `infrastructure.validation` (84 files)

Markdown + PDF + integrity CLIs underpinning Stage 04 diagnostics.

6.1.23 `infrastructure/logrotate.d` (0 files)

Operational templates for deployments (documentation-first; intentionally minimal Python footprint).

Documentation maturity: Coverage statements in Results pull from introspection—not hand-maintained denominators—so newly promoted modules automatically flow into manuscripts after `generate_manuscript_metrics.py`.
FAIR+RSE linkage: MCP-ready `SKILL.md` artefacts align with evaluator heuristics (executability + metadata richness) emphasized by FAIRsoft guidance [Garijo et al., 2024].

7 Security and Provenance

Research integrity requires more than reproducibility; it requires verifiable authorship. In an era of generative AI, automated scraping, and synthetic media, the ability to prove that a document was produced by a specific pipeline at a specific time is a critical defense against fabrication and misattribution. The W3C PROV data model [Moreau and Missier, 2013] establishes a formal vocabulary for expressing provenance records—entities, activities, and agents connected by derivation, generation, and attribution relations. Digital watermarking, pioneered by Cox et al. [Cox et al., 1997] for multimedia integrity verification, provides the foundational signal-processing theory for embedding imperceptible provenance markers within artifacts. `template/` implements a domain-specific provenance layer that embeds these relations directly into the PDF artifact itself, using four complementary steganographic and cryptographic mechanisms.

7.1 Threat Model

The steganography subsystem defends against three classes of threats:

1. **Unauthorized redistribution**: A manuscript is scraped and republished without attribution. The steganographic watermark survives the redistribution and can be used to prove original authorship.
2. **Content tampering**: Figures or text are modified after publication. The SHA-256 hash embedded in the PDF metadata detects any modification to the file contents.
3. **Provenance forgery**: An attacker claims to have produced a document they did not author. The build timestamp, commit hash, and pipeline metadata embedded in the watermark provide a verifiable chain of custody.

7.2 Steganographic Layers

The system applies four complementary layers of provenance information:

7.2.1 Layer 1: PDF Metadata Injection

The `inject_pdf_metadata` function writes structured metadata into both the PDF Info dictionary and an XMP (Extensible Metadata Platform) packet:

- `/Creator`: Pipeline identifier
- `/Producer`: Module path (`infrastructure.steganography`)
- `/CreationDate`: UTC timestamp in ISO 8601 format
- `/Author`: From `config.yaml`
- `/Title`: From `config.yaml`
- Custom fields: DOI, ORCID, repository URL

7.2.2 Layer 2: Cryptographic Hashing

Before watermarking, a SHA-256 hash of the rendered PDF is computed and stored in:

- The output manifest (`output/manifest.json`)
- The PDF metadata (`/Subject` field)
- An external hash file (`output/<name>.sha256`)

This enables post-hoc verification: anyone with the hash can verify that the PDF has not been modified since rendering.

7.2.3 Layer 3: Alpha-Channel Text Overlay

A semi-transparent text overlay is applied to each page of the PDF, encoding:

- Build timestamp
- Git commit hash (short SHA)
- Project name
- Pipeline version

The overlay is rendered at low opacity (typically 3–5% alpha) to be invisible during normal viewing but detectable through image analysis. It survives printing (as a faint watermark) and standard PDF operations. A representative overlay text string takes the following form:
`template/ | built: 2026-03-19T14:23:11Z | commit: a4f2c1b | pipeline: v2.0.0 | project: template`
This single line, tiled across each page at 3–5% opacity, encodes the complete build provenance chain: the system identifier, ISO 8601 build timestamp, short Git commit hash, pipeline version, and project name. Together these fields allow a verifier to reconstruct—from the watermark alone—which version of the code, at which moment in time, produced the document.

7.2.4 Layer 4: QR Code Injection

An optional QR code is generated containing a URL pointing to the repository (e.g., `github.com/docxology/template`). The QR code is placed in a configurable position (default: bottom-right corner of the last page) at a specified size.

7.3 The `secure_run.sh` Orchestrator

The steganographic pipeline is orchestrated by `secure_run.sh`, a Bash script that wraps the standard `run.sh` pipeline with post-processing steganography:

1. Execute the standard YAML-declared pipeline (12 stages; default full 10) pipeline for the target project.
2. The `secure_run.sh` script invokes `SteganographyProcessor`.
3. Apply metadata injection, hashing, text overlay, and QR code injection.
4. Save the secured PDF alongside the original.
5. Generate a steganography report in JSON format.

The orchestrator processes either a single specified project or all discovered projects sequentially.

7.4 Tamper Detection

Verification is performed by comparing the stored SHA-256 hash against a freshly computed hash of the distributed PDF. Any modification—even a single bit flip—produces a hash mismatch. The alpha-channel overlay provides a secondary, visual verification channel that does not require access to the original hash.

7.5 Limitations

- Alpha-channel overlays are stripped by some PDF optimization tools (e.g., `qpdf --optimize`).
- QR codes are visible and may be removed by a determined attacker.
- The system does not provide non-repudiation in the cryptographic sense—it does not use digital signatures with private keys. Future versions may integrate GPG or X.509 signing.
- The provenance model is pipeline-centric rather than fully PROV-compliant. The path from `template/`'s current metadata-based provenance to full W3C PROV-compliant traces involves four steps: (1) *entity identification*—assigning stable identifiers (URIs or SWHIDs) to each pipeline input (manuscript files, data, config); (2) *activity logging*—recording each pipeline stage as a PROV Activity with start/end timestamps (already encoded in the watermark overlay); (3) *agent attribution*—binding each Activity to the pipeline version and Git commit hash (already encoded in the overlay and PDF metadata); (4) *PROV-O serialization*—emitting the provenance graph as OWL-RDF (PROV-O) or text (PROV-N) alongside the PDF. Steps (2) and (3) are already implemented; steps (1) and (4) are the primary remaining gaps. A future `infrastructure.provenance` module would close both gaps automatically.

7.6 Relationship to Software Supply Chain Integrity

The steganographic provenance layer operates at the *document level*—it certifies the integrity of a specific PDF artifact. A complementary concern is *build-level* provenance: certifying that the pipeline itself was executed with verified source code and dependencies. Frameworks such as in-toto [Torres-Arias et al., 2019] and SLSA (Supply-chain Levels for Software Artifacts) address this concern by defining attestation chains

from source commit through build steps to final artifact. The NTIA's minimum elements for a Software Bill of Materials (SBOM) [National Telecommunications and Information Administration, 2021] further standardize the enumeration of software components and dependencies—essential for establishing the provenance lineage of build environments. SLSA defines four graduated levels of build integrity:

SLSA Level	Requirement	`template/` Status
1	Provenance document exists	✓ SHA-256 manifest + steganographic metadata
2	Version-controlled build scripts	✓ All scripts in git
3	Isolated build environment	about Docker support exists but not enforced in CI
4	Hermetic, reproducible builds	N – future work

Future versions of `template/` may generate SLSA-compatible provenance attestations alongside the steganographic watermarks, creating a two-layer provenance model: in-toto attests that the build pipeline was executed with the claimed source code, while the steganographic layer attests that the PDF was produced by that pipeline at a specific time.

7.7 Relationship to FAIR and Formal Provenance Standards

The steganographic layer supports the FAIR for Research Software (FAIR4RS) principles [Barker et al., 2022] at the artifact level. PDFs carry embedded metadata (Findability) in standardized XMP format (Interoperability). The SHA-256 hash manifest enables persistent integrity verification (a prerequisite for Reusability). The Documentation Duality standard ensures that the software producing the artifact is inspectable and well-documented (satisfying FAIRsoft [Garijo et al., 2024] metadata and documentation indicators). Full PROV-compliant provenance traces—capturing the derivation chain from source data through analysis scripts to rendered PDF—are a natural extension and a primary target for future development.

Software Heritage [Di Cosmo et al., 2020] complements this picture at the source-code level: by archiving the `template/` repository and assigning a reproducible SWHID (Software Hash Identifier) to each commit, Software Heritage makes the pipeline itself—not just its output—a citable, persistent digital artifact. A published SWHID alongside the PDF DOI creates a complete, two-artifact citation record: the paper's content is versioned via DOI; the code that generated it is versioned via SWHID. This combination satisfies the Katz et al. [Katz et al., 2021] software citation principles' requirement that software used in research be independently citable and permanently accessible.

8 Appendices

8.1 Appendix: Pipeline Stage Reference

Table 1: Single-project DAG exported from default `pipeline.yaml` (names shown in topological order). Scripts live under `scripts/`.

Stage name	Script / method	Primary inputs	Outputs / artefacts	Failure mode
Clean Output Directories	`_run_clean_outputs`	prior `projects/<name>/output/`, mirrored `output/<name>/` targets	emptied trees	Blocking
Environment Setup	`00_setup_environment.py`	toolchain probes	scaffold dirs, env exports	Blocking
Infrastructure Tests	`01_run_tests.py --infra-only --infra-scope pipeline-smoke`	`tests/infra_tests/`	coverage + junit-style logs	tolerant ceilings
Project Tests	`01_run_tests.py --project-only`	`projects/<name>/tests/`	coverage artefacts	blocking by default
Project Analysis	`02_run_analysis.py`	thin scripts	`figures/`, `data/`, reports	Blocking
PDF Rendering	`03_render_pdf.py`	`manuscript/`, placeholders	`.pdf`/`.tex` bundles	Blocking
Output Validation	`04_validate_output.py`	render tree	Markdown + PDF diagnostics JSON	Blocking / downgrade
LLM Scientific Review	`06_llm_review.py --reviews-only`	resolved manuscript artefacts	textual reviews	Optional skip (`allow_skip`)
LLM Translations	`06_llm_review.py --translations-only`	abstract metadata	multilingual snippets	Optional skip (`allow_skip`)
Copy Outputs	`05_copy_outputs.py`	validated tree	mirrored `output/<name>/…`	soft fail logged
Executable Bundle	`08_executable_bundle.py`	project tree + outputs	container bundle manifest	opt-in (`bundle` tag)
Archival Publication	`09_archive_publication.py`	bundle + deliverables	archival deposit manifest	opt-in (`archival` tag)

`scripts/07_generate_executive_report.py` is invoked **outside** this DAG whenever `execute_multi_project.py` aggregates pipelines—supplying cross-project KPI dashboards absent from lone-project checkpoints.

8.2 Appendix: Configuration Reference

Table 2: Configuration schema for `config.yaml`, showing all supported fields and their structure.

```yaml
paper:
  title: "Paper Title"
  subtitle: "Optional Subtitle"
  version: "1.0"
  date: "2026-03-19"

authors:
  - name: "Author Name"
    orcid: "0000-0000-0000-0000"
    email: "author@example.com"
    affiliation: "Institution"
    corresponding: true

publication:
  doi: "10.5281/zenodo.XXXXXX"
  journal: "Target Journal"
  volume: "1"
  pages: "1-10"
  year: "2026"

keywords:
  - "keyword1"
  - "keyword2"

metadata:
  license: "Apache License 2.0"
  language: "en"

llm:
  reviews:
    enabled: true
    types: [executive_summary, quality_review]
  translations:
    enabled: false

testing:
  max_test_failures: 0
  max_infra_test_failures: 3
  max_project_test_failures: 0
```

8.3 Appendix: Repository Directory Structure

```
template/
  infrastructure/
     config/ docker/ documentation/ llm/
     orchestration/   # Thin Python entry equal to `./run.sh` backend
     prose/ reference/ rendering/ reporting/
     scientific/ search/ skills/ steganography/ validation/
     project/ core/
     logrotate.d/      # Operational rotation templates (no Python pkg)
  scripts/
     00_setup_environment.py … 07_generate_executive_report.py
     execute_pipeline.py execute_multi_project.py
  projects/                    # Typed program subfolders (`discover_projects`)
     templates/               # Public exemplars (git-tracked)
        template_active_inference/
        template_autoresearch_project/
        template_code_project/
        template_prose_project/
        template_template/   # Present manuscript (`manuscript/` here)
     active/                   # Hot-seat rendered set (symlinked, private)
     working/                  # Non-rendered backburner (symlinked, private)
     published/                # Non-rendered published (symlinked, private)
     archive/                  # Non-rendered retired (symlinked, private)
     other/                    # Non-rendered misc (symlinked, private)
  docs/ (17 top-level areas, 378+ markdown files per live counter)
  tests/                        # Infra suites (467+ files)
  AGENTS.md / README.md / CLAUDE.md / pyproject.toml
  run.sh / secure_run.sh
  output/ …                     # Mirrors after copy stage
```

See `docs/_generated/active_projects.md` for regenerated slugs (`uv run python scripts/generate_active_projects_doc.py`).

8.4 Appendix: Exemplar Project Summary

Table 3: Three representative workspaces under `projects/templates/` illustrating heterogeneous domains while sharing Layer 1. This is a hand-picked sample, not the full roster: the complete public exemplar set (currently nine workspaces) is enumerated dynamically from `PUBLIC_PROJECT_NAMES` and listed below.

The full public exemplar roster is: `templates/template_active_inference`, `templates/template_autoresearch_project`, `templates/template_autoscientists`, `templates/template_code_project`, `templates/template_eda_notebook`, `templates/template_gold_refinement`, `templates/template_literature_meta_analysis`, `templates/template_madlib`, `templates/template_methods_paper`, `templates/template_newspaper`, `templates/template_prose_project`, `templates/template_search_project`, `templates/template_sia`, `templates/template_template`, `templates/template_textbook`. The three rows below are a representative sample; a future `exemplar_summary_table` token in `build_manuscript_metrics_dict` would let this table cover every exemplar without hand-editing.

Project slug	Purpose	Highlights	Tests	Figures (Stage 02 hint)
`template_code_project`	Optimization tutorial	Convex demo figures, scripted tables	236 @ 90%+ gate	Controlled matplotlib exports
`template_prose_project`	Prose-heavy workflow	Validates narrative-only repos	120	Lightweight / optional plots
`template_autoresearch_project`	AutoResearch readiness	Planner + validation CLI	296	Readiness reports from Stage 02

Meta manuscript location: introspective study lives in `projects/templates/template_template/` beside the public exemplar set. Discovery now follows the typed `projects/` layout—`projects/templates/**` and `projects/active/**` are discovered/rendered, while `projects/working/**`, `projects/published/**`, `projects/archive/**`, and `projects/other/**` remain non-rendered—see root `CLAUDE.md` for invocation patterns (`resolve_project_root`).

8.5 Appendix: Documentation Inventory

The repository maintains documentation at three levels:

Table 4: Documentation inventory across the four-layer documentation architecture, from repository-wide system files to per-module skill descriptors.

Level	Files	Purpose
Repository root	`AGENTS.md`, `CLAUDE.md`, `README.md`, `RUN_GUIDE.md`	Global navigation and AI agent context
`docs/` directory	378 files across 17 subdirectories	User guides, API reference, troubleshooting
Per-directory	`AGENTS.md` + `README.md` at every directory	Documentation Duality standard
Per-module (Tier 3)	`SKILL.md` at every infrastructure module	Machine-parseable MCP-aligned skill descriptor
Infrastructure-level (PAI)	`PAI.md` at `infrastructure/` directory	Personal AI Infrastructure integration contract

The `docs/` subdirectories cover: `core/` (essential docs), `guides/` (skill levels 1–12), `architecture/` (system design), `usage/` (content authoring), `operational/` (build, config, logging, troubleshooting), `reference/` (API, FAQ, glossary), `modules/` (23 infrastructure modules), `development/` (contributing, testing), `best-practices/` (version control, migration), `prompts/` (21 AI prompt templates), `security/` (steganography, hashing), and `audit/` (review reports).

Every count in this appendix is injected from live repository introspection rather than hand-maintained: `378` counts every Markdown file beneath `docs/` recursively, `17` counts its first-level subdirectories, and `21` counts the workflow subdirectories that each carry a `SKILL.md` descriptor. This is the same discipline the manuscript argues for throughout—a hand-typed "90+ files across 12 subdirectories" silently rots as the tree grows, whereas a token re-resolves on every render. A reader onboarding to the repository should start at `docs/core/`, follow the graduated `docs/guides/` skill ladder, and consult the per-directory `AGENTS.md`/`README.md` pair nearest to whatever code they are editing; AI agents additionally read each module's `SKILL.md` to locate capabilities without guessing API signatures.

8.6 Appendix: Comparative Tool Matrix

Symbol key (applies to all cells): **Y** = full native support · **about** = partial or plugin-based · **N** = absent. See also Figure 4 for a colour-coded heatmap rendering of this table.

Table 5: Comparative feature matrix (14 capabilities × 10 tools). Y = full native support, about = partial or plugin-based, N = absent.

Capability	`template/`	Snakemake 9	Nextflow 25	CWL 1.2	Quarto 1	Jupyter Book 2	R Markdown	DVC 3	Overleaf (2025)	OpenAI Prism
Pipeline orchestration	Y	Y	Y	Y	about	N	N	Y	N	N
Manuscript rendering	Y	N	N	N	Y	Y	Y	N	Y	Y
Testing enforcement	Y	N	N	N	N	N	N	N	N	N
Coverage thresholds	Y	N	N	N	N	N	N	N	N	N
Cryptographic provenance	Y	N	about [1]	N	N	N	N	about [2]	N	N
Steganographic watermarking	Y	N	N	N	N	N	N	N	N	N
Multi-project management	Y	N	N	N	N	N	N	N	about	about
AI-agent documentation	Y	N	N	N	N	N	N	N	about	about
Agentic skill protocol (SKILL.md / MCP)	Y	N	N	N	N	N	N	N	N	N
Interactive TUI	Y	N	N	N	N	N	N	N	N	N
Zero-mock policy	Y	N	N	N	N	N	N	N	N	N
Container support	N	Y	Y	Y	N	N	N	N	N	N
Distributed execution	N	Y	Y	Y	N	N	N	about [3]	N	N
Multi-language (R/Julia)	N	Y	N	Y	Y	Y	Y	Y	N	N

[1] Nextflow 25.04.0 introduced data-lineage provenance tracking (build-level, not document-level). [2] DVC provides content-addressed versioning for data artifacts via its object store. [3] DVC integrates with remote

storage (S3, GCS, Azure) but does not natively orchestrate distributed compute. [4] Overleaf and OpenAI Prism are collaborative cloud LaTeX/AI writing environments; their AI features (GPT-5.2 for Prism, Overleaf Labs AI for Overleaf) are partial/early-stage as of 2025–2026.

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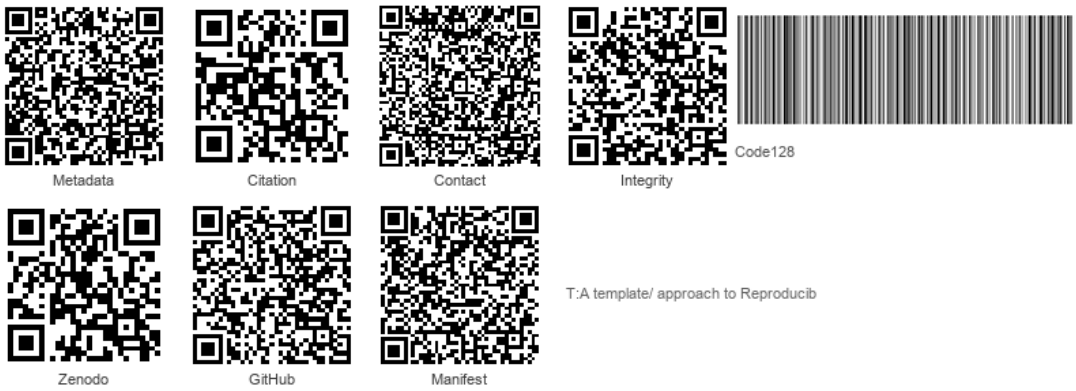

Figure 3: Integrity QR strip

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